

Place-based policies

EC3335 Set 09

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Content

(i) Place-based policies

(ii) Utility framework

Housekeeping

PS03 (*due Mon, Nov. 21*)

- Posted now (a couple of minutes ago)

PS04 (*due Wed, Nov. 30*)

- will be posted by Mon next week

TotC book report (*due Thu, Dec. 8*)

- Short (1000 words) assignment
- rubric posted on canvas and on the course repo
- will be checking for plagiarism

Final (*14:45 Wed, Dec. 7*)

- final schedule
- Comprehensive exam with an emphasis on the newer material

Place-based policies

Place-based policies

Definition: Place-based policies

- Location specific policies/laws

What are some examples?

- State and Local Taxes
- State/City minimum wage
- Abortion restrictions
- Air quality monitoring
- Zoning & Land Use Restrictions
- Enterprise Zones
- Medicinal and recreational marijuana laws

Federal policies that are **uniform across all states** *are not* place-based

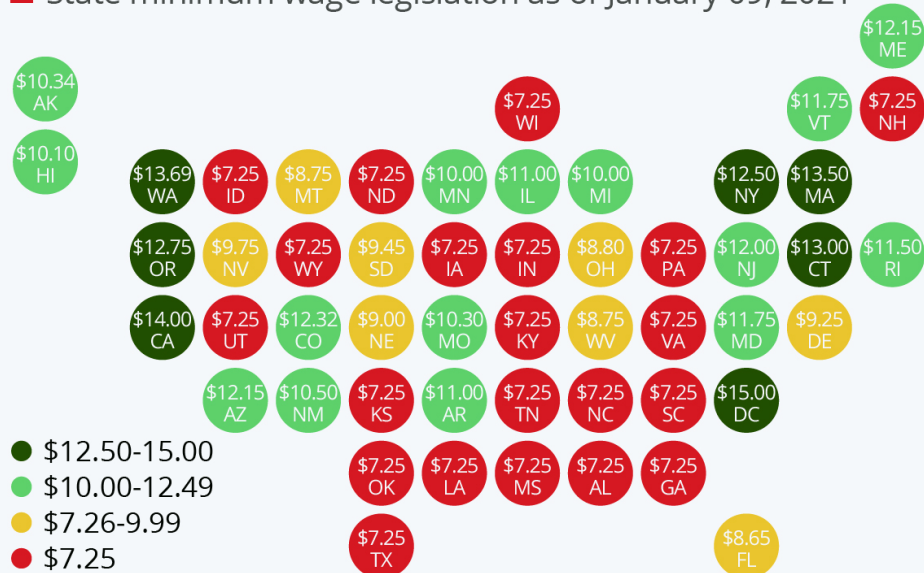
- Harder to migrate across **international borders** than state borders

Place-based policies: Examples

Pb polcies: Minimum wage

The U.S. Minimum Wage By State

State minimum wage legislation as of January 09, 2021*



* Alabama, Louisiana, Mississippi, South Carolina and Tennessee have not adopted a minimum wage while Georgia and Wyoming are below the \$7.25 federal minimum. In all of these states, the federal minimum applies.

Source: National Conference of State Legislatures



Federal Minimum Wage: 7.25 (not a place based policy)

Pb polcies: Enterprise zones

Definition: Enterprise zone:

A geographic area that has been granted **tax breaks, regulatory exemptions, or other public assistance** in order to encourage private economic development and job creation

Examples:

- Jersey City, NJ since 1983
- China: Shanghai and Shenzen (Special Economic Zones (SEZ))

Pb polcies: Brownfield remediation

Definition: Brownfield:

A geographic area that has previously been developed land that is not currently in use due to industrial and/or commercial pollution

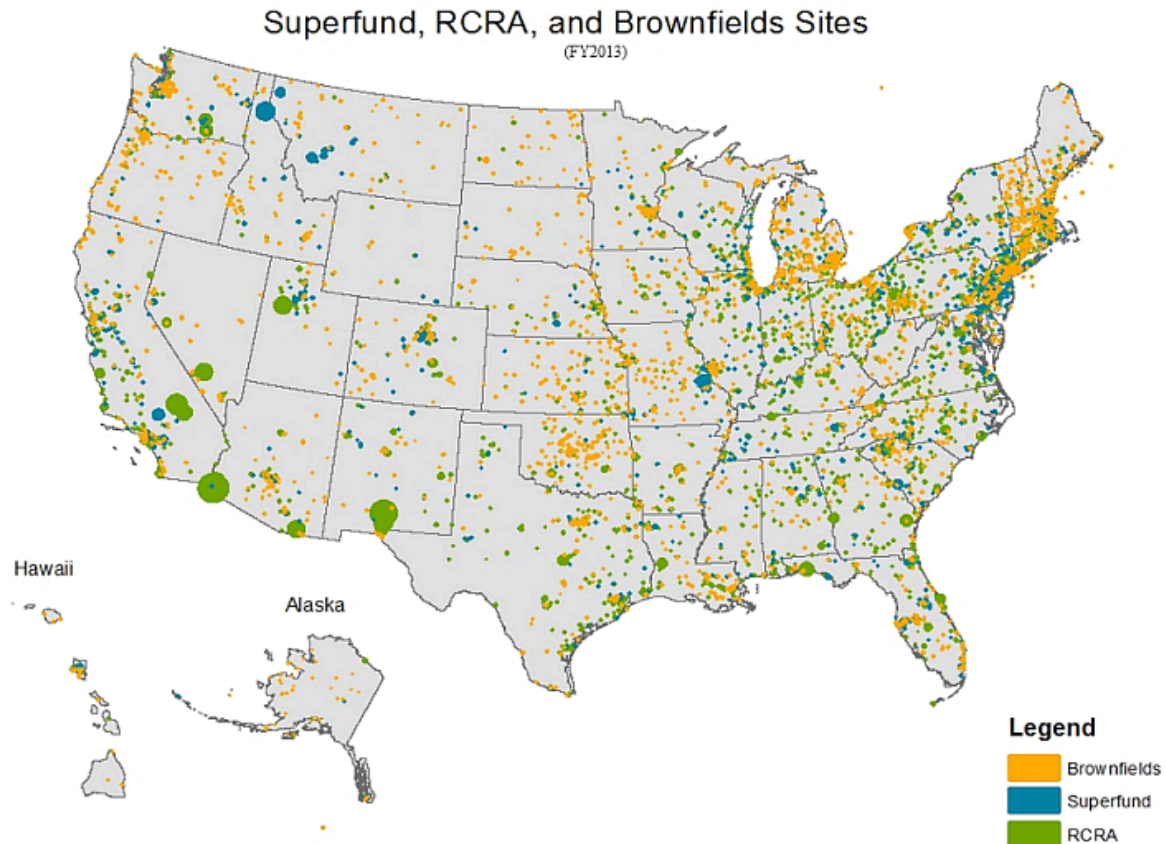
Examples include abandoned business such as:

- Gas stations
- Dry cleaning
- Factories
- Mills
- Foundries

There are several Brownfields in the Eugene/Springfield area

- Ninkasi over took a brownfield to expand brewing operations in 2012

Pb polcies: Brownfields remediation



Pb polcies: Brownfields remediation

Brownfields lower the amenity value of neighborhoods

- High health costs associated with living near a brownfield
 - Petroleum leaks from underground storage tanks lead to increases in the probability of low birth weights and preterm birth by **7-8 percent**
- Tremendously expensive to clean up

Land is not used it is not contributing to local economies- opportunity cost

Cleaning these up raises **amenity value** of the neighborhood

Property values around brownfields are far lower than comparable land

- **Gentrification**

Pb policies: Air quality

December 2, 1970: Environmental Protection Agency (EPA) is Established

- Included the Clean Air Act
 - Regulates county level air quality with a system of air monitors

Following years: amendments to the CAA

- **1990:** Additional power granted to state/local authorities to enforce air quality standards
- **1997:** PM 2.5 (particulate matter of 2.5 micrograms or less) standards placed
- **2005:** PM2.5 standards enforced
- **2011:** Standards for greenhouse gases

Pb polcies: Air quality

Particulate Matter (**PM**) in the US is regulated at the **county level**[†]

If a county exceeds certain threshold for **PM** , all firms over a certain size need to pay a pretty big fine

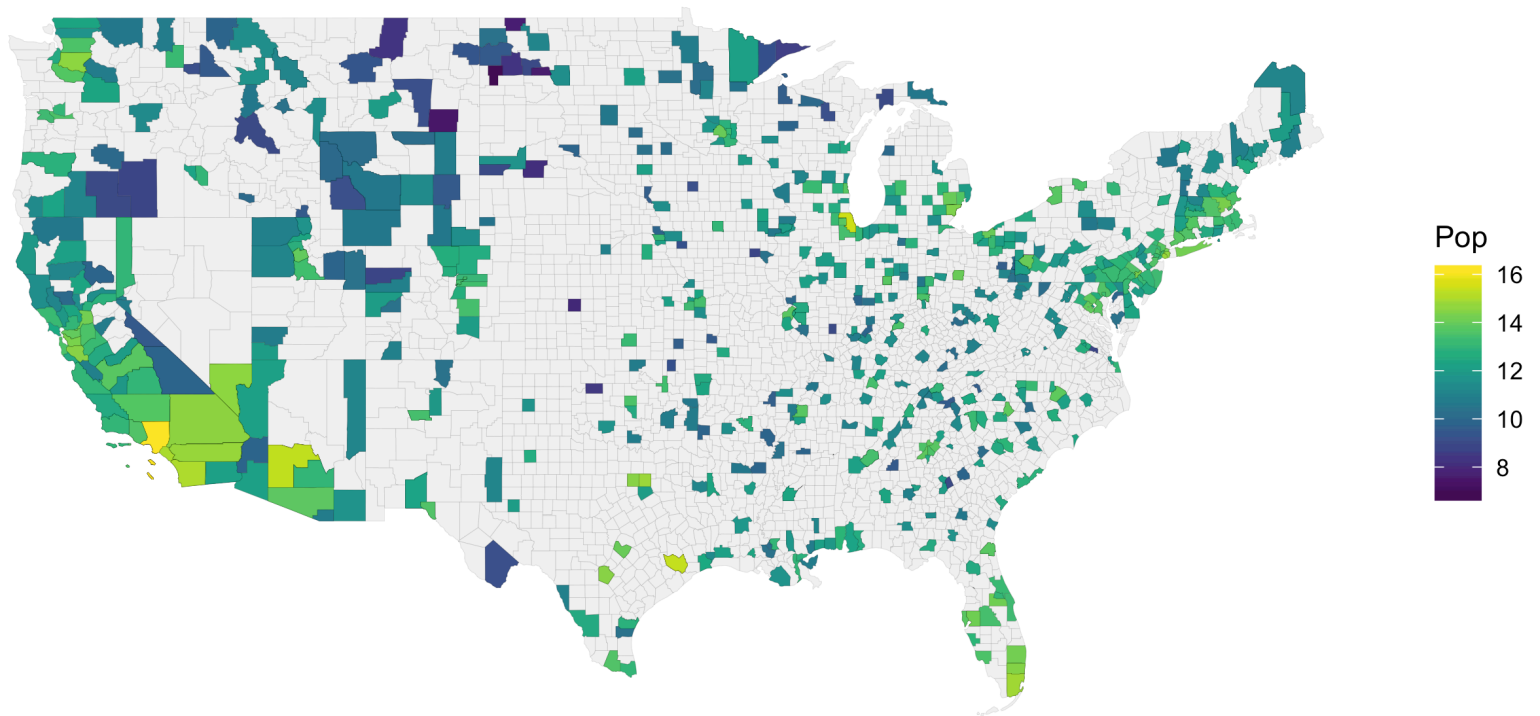
- Exceptions for fires, other natural events

[†] For more details, look [here](#)

Pb polcies: Air quality

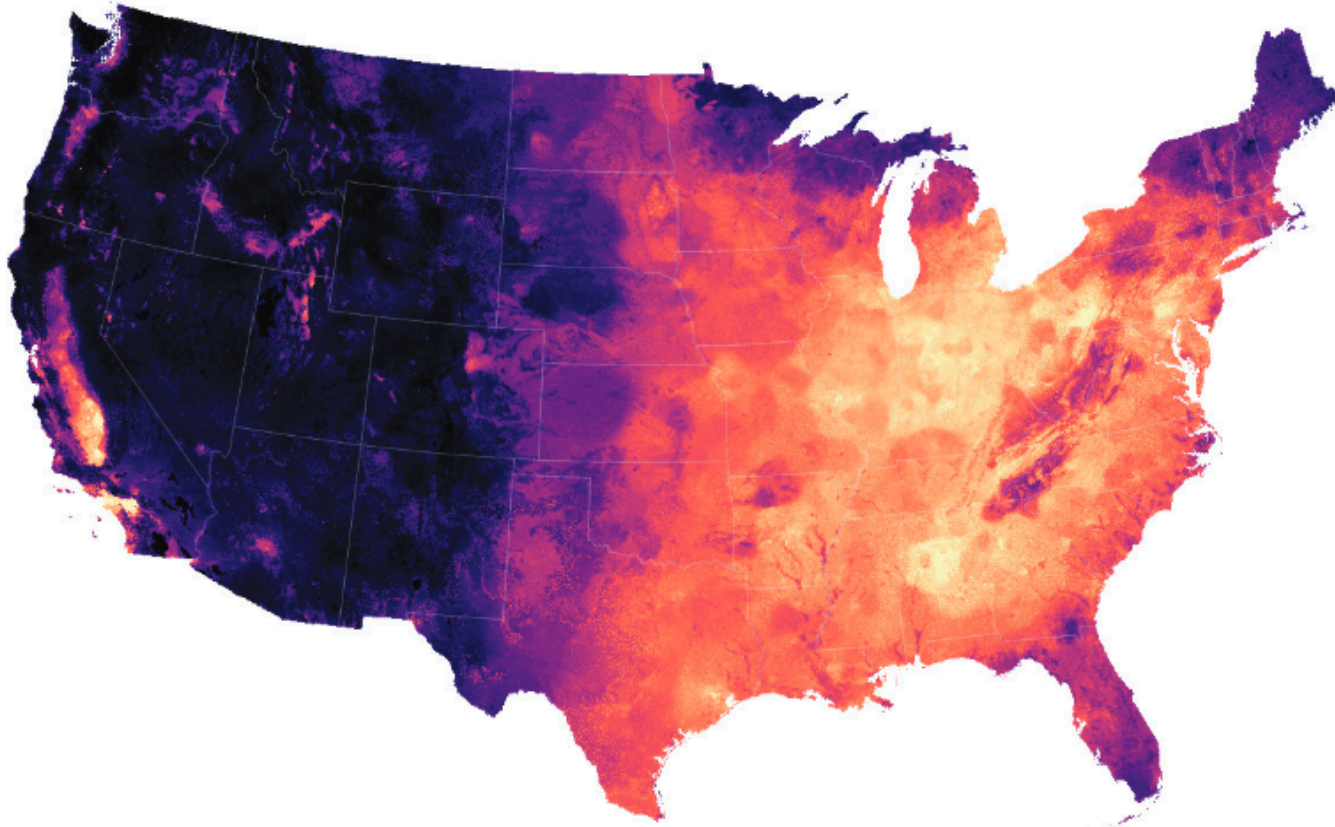
US Counties with PM2.5 Monitors

Population measured in logs



Sources: US EPA and Census

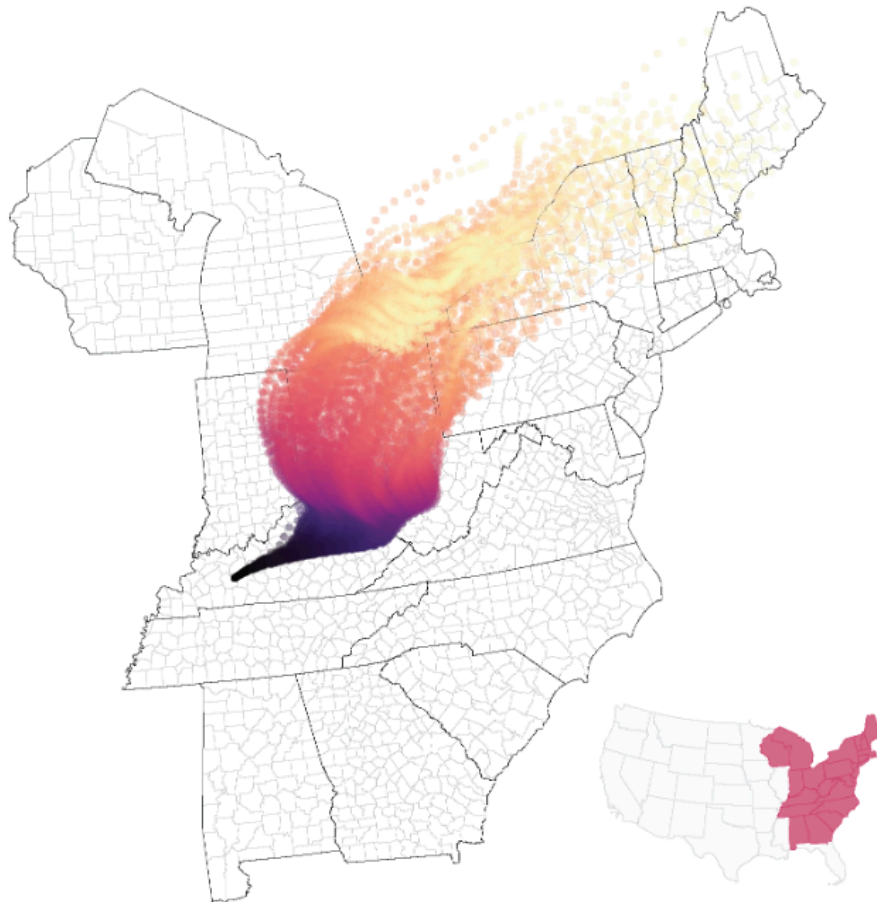
Pb polcies: Air quality



- Di et al. (2016)

Pb polcies: Air quality

Does air quality monitoring make sense at a local level? Why or why not?



Pb polcies

Why do we care about place-based policies?

People are **mobile** and **respond to changes in incentives**

Place-based policies influence **location decisions**

- TotC give really good intuition in the chapter about Detroit.

Question: Why do federal policies impact cities differently?

Min wage: might be **binding** in some states, others not

- Some labor markets might be competitive. Others not
- Federal Income Tax: Cost of Living varies by state.

Utility framework

Utility framework

This next section of the class will add another layer of complexity

Set up a **utility framework** to understand how policies impact welfare --
Only scratches the surface of how one may model impacts of pb policy

Some of these examples are based on [Mark Colas'](#) notes

- Learn more about this in his 400 urban economics class

Utility framework

Utility: Abstract notions of people's preferences.

- Location based policies impact individual **location decisions**

Model decision through the lens of an individual's utility (welfare)

Suppose: $U(\text{City A}) > U(\text{City B}) \implies$ Moving to City A

Suppose City B makes a policy change that raises wages:

- $U(\text{City A}) < U(\text{City B}) \implies$ Moving to City B

Changes in a location based policy are going to change **incentives**

- eg. San Diego has extremely strict zoning restrictions

Zoning restrictions $\implies$ limited housing supply $\implies$ high rents $\implies$ "why do i live here.." $\implies$ move to Oregon

Utility framework

Example: Preferences (Leontief) over left-shoes and right-shoes may be expressed with the following utility function:

$$U(\text{left shoes}, \text{right shoes}) = \min \{\text{left shoes}, \text{right shoes}\}$$

Q: In words, what does this say?

A: Another right shoe does nothing for me unless I get another left shoe

Q: Give the above utility function, which bundle would I rather consume?

bundle 1 : (10000, 1)

bundle 2 : (2, 2)

A: $U(10000, 1) = 1 < U(2, 2) = 2$, so I would rather consume bundle 2

Utility framework

Main point: Utility is used to rank outcomes

Remember: Utility is **ordinal** *not* **cardinal**

This means: we can only speak to the ordering of outcomes, not the levels

- Many utility functions give equivalent preference rankings

What if utility over shoes was:

$$U_2(\text{left shoes}, \text{right shoes}) = 10 * \min \{\text{left shoes}, \text{right shoes}\}$$

Q: Does this represent the same underlying preferences as before?

A: Yes, because $U_2(10000, 1) = 10 * 1 = 10 < U_2(2, 2) = 10 * 2 = 20$

- So the bundle (2, 2) is still preferred to (10000, 1)

Utility framework: Locations

Could we write a **utility** function over **locations**? **Yep!**

What do people make location decisions on?

*What would a locational utility function take as **inputs**?*

For now, assume people only care about **3** features of locations:

wages, **rents**, **amenities**

These all vary across locations, right? (first part of this class)

Let w_j , r_j , and a_j denote wages, rents, and amenities in location j

Utility framework: Locations

Let w_j , r_j , and a_j denote wages, rents, and amenities in location j

- $j = SF$, for example

General form: $U(w_j, r_j, a_j) = U_j$

- This says utility in location j is a function of wages, rents, and amenities, in location j

In practice, could write down an infinite number of functions for $U(\cdot)$

Usual assumptions:

- Higher wages are better
- Lower rents are better
- More amenities are better

Utility framework: Locations

Example: Assume linear utility functions and everyone is identical:

$$U(w_j, r_j, a_j) = w_j - .5 * r_j + a_j$$

Suppose our two locations are SF and OAK again. If:

- $w_{SF} = 10, r_{SF} = 8, a_{SF} = 4$
- $w_{OAK} = 8, r_{OAK} = 3, a_{OAK} = 1$

Q: How do workers sort across the cities?

- $U(w_{SF}, r_{SF}, a_{SF}) = 10 - .5 * 8 + 4 = 10$
- $U(w_{OAK}, r_{OAK}, a_{OAK}) = 8 - .5 * 3 + 1 = 7.5$

Well $10 > 7.5$ so... everyone moves to SF

Utility framework: Locations

Is it **reasonable** that everyone would move to SF? What are we missing?

Was that last example a locational equilibrium by definition? **Nope**

In **locational equilibrium**, utility is **equalized across locations**

Can't have: $U(w_{SF}, r_{SF}, a_{SF}) > U(w_{OAK}, r_{OAK}, a_{OAK})$

How can **"fix up"** our last example such that we model locational eq?

We can **allow rents** (or wages or both) **to adjust** such that utility is equivalent across the two cities

Utility framework: Locations

Another Problem: People move and utility is equal across all locations

Thus far, we assume **wages** and **rents** are exogenous

*This is a **bad assumption** right?*

Let rents, but not wages, adjust to individual location decisions

- (ie make rents **endogenous** to the model)

Utility framework: Rents

Rents in every city given by:

$$r_j(L_j) = 2 \times L_j$$

- $r_j(L_j)$: rents *are a function* of the population (not multiplied)
- L_j is the pop in city j ; choosing 2 was arbitrary

Suppose we have two cities 1 and 2, with 7 people total: $L_1 + L_2 = 7$

Utility: $U(w_j, r_j(L_j), a_j) = w_j - .5 \times r_j(L_j) + a_j$

Wages: $w_1 = 12, w_2 = 7$

Rents: $r_j(L_j) = 2 * L_j$

Amenities: $a_1 = a_2 = 0$

Utility framework: Rents example

Suppose we have two cities 1 and 2, with 7 people total: $L_1 + L_2 = 7$

Utility: $U(w_j, r_j(L_j), a_j) = w_j - .5 \times r_j(L_j) + a_j$

Wages: $w_1 = 12, w_2 = 7$

Rents: $r_j(L_j) = 2 * L_j$

Amenities: $a_1 = a_2 = 0$

Qs: How many people live in each city? What are rents in each city?

Note: You have **two equations** and **two unknowns** (namely, L_1 and L_2)

- $U(w_1, r_1(L_1), a_1) = U(w_2, r_2(L_1), a_2)$ (from locational eq)
- $L_1 + L_2 = 7$ you know the total population

Utility framework: Rents example

Locational eq gives:

$$\begin{aligned}w_1 - .5 * r_1(L_1) &= w_2 - .5 * r_1(L_2) \\12 - .5 * (2 * L_1) &= 7 - .5 * (2 * L_2) \\-L_1 &= -5 - L_2 \\L_1 &= 5 + L_2\end{aligned}$$

Population must sum to 7. Thus:

$$\begin{aligned}L_1 + L_2 &= 7 \\5 + L_2 + L_2 &= 7 \\2 * L_2 &= 2 \\L_2 = 1 &\implies L_1 = 6\end{aligned}$$

Utility framework: Place based policies

Ok, how do we tie this back into **place-based** policies?

Example

Initial equilibrium: $U(w_j, r_j(L_j), a_j) = k$ for all cities j

Suppose SF implements a 30%, flat, income tax

- Post-tax wage in city SF is now $w_{SF}^{tax} = 0.7 * w_{SF}$
- Assume **wages are fixed**, but **rents adjust to population**

Utility in city j is:

$$U(w_{SF}^{tax}, r_{SF}(L_{SF}), a_{SF}) < U(w_{SF}, r_{SF}(L_{SF}), a_{SF})$$

If utility is **increasing in wages**, then an income-tax lowers utility.

Utility framework: Equilibrium

Can it be an equilibrium if:

$$U(w_{SF}^{tax}, r_{SF}(L_{SF}), a_{SF}) < U(w_{SF}, r_{SF}(L_{SF}), a_{SF})$$

No!

Because $U(w_{SF}, r_{SF}(L_{SF}), a_{SF}) = k$

So $U(w_{SF}^{tax}, r_{SF}(L_{SF}), a_{SF}) \neq k$

Thus people move **away from SF** and rents fall

So utility goes up in SF until $U(w_{SF}^{tax}, r_{SF}(L_{SF}), a_{SF}) = k$

Extensions

This flexible way of modeling gives us many options for modeling place based policies

- Other kind of subsidies/taxes: goes into w_j
- Rent subsidies or property taxes: impacts r_j
- **Q:** How would you model an increase in public school quality?

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